

Thesis Defense:
Invariant measures for the stochastic heat equation with
no drift term

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The stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x), & t \geq 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

- The Stochastic Heat Equation is interpreted as follows:

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)\mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)b(u(s, y))W(ds, dy)$$

where $G(t, x)$ is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

- A processes, $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$, that satisfies this relation is referred to as the *mild solution*.

The noise term $\dot{W}$

- $W = \{W(\phi) : \phi \in \mathcal{S}(\mathbb{R}^d)\}$ is a centered Gaussian process uniquely defined through the following covariance functional:

$$J(\phi, \psi) := \mathbb{E}[W(\phi)W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where $\tilde{\psi}(s, x) = \psi(s, -x)$. We have the following nomenclature.

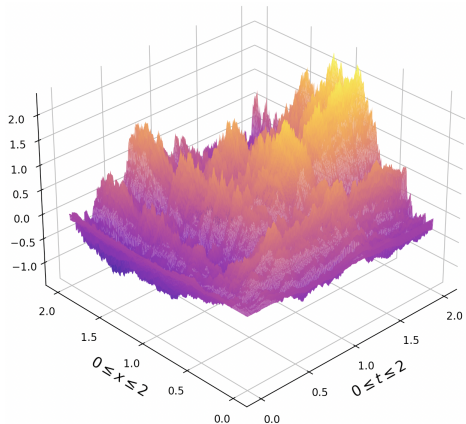
- *Space-time white noise*: $\Gamma(dx) = \delta_0(dx)$.
- *Time white and spatially colored*: $\Gamma(dx) = f(x)dx$ where f is a non-negative and non-negative definite function referred to as the *Correlation function* and its Fourier transform,

$$\hat{f}(\xi) := \mathcal{F}f(\xi) = \int_{\mathbb{R}^d} \exp(-ix \cdot \xi) f(x)dx,$$

is referred to as the *spectral density*.

Proposition (DaPrato and Zabczyk [7, Propsition 4.19])

A space-time white noise, dB , is the distributional derivative of a Brownian sheet, $B(t, x)$.



White noise expansion DaPrato [7]

If $t \geq 0$ and $0 \leq x \leq \pi$ then,

$$B(t, x) = \sum_{n=0}^{\infty} \frac{\beta_n(t)}{n + \frac{1}{2}} \sqrt{\frac{2}{\pi}} \sin \left[\left(n + \frac{1}{2} \right) x \right]$$

Walsh [8]

$$\int_t^{t+\delta_t} \int_x^{x+\delta_x} dB(ds, dy) \sim N(0, \delta_t \delta_x)$$

Brownian Sheet Simulation (Walsh [8]):

$$B(t, x) \sim B(t - \delta_t, x) + B(t, x - \delta_x) - B(t - \delta_t, x - \delta_x) + N(0, \delta_t \delta_x)$$
$$B(t, 0) = B(0, x) = 0 \text{ for all } t, x > 0.$$

Existence and uniqueness with space-time white noise

- A necessary condition for global existence, *Dalang's condition* [6], is the following:

$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) < \infty, \quad \forall t > 0.$$

- Direct calculations yields that,

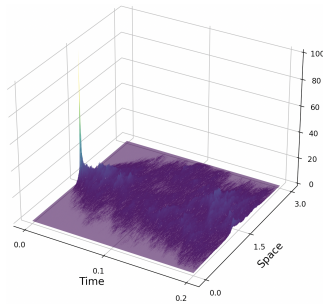
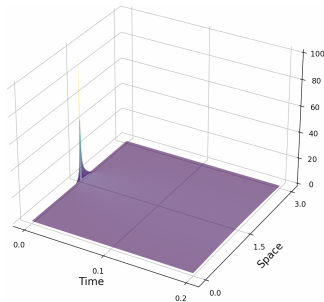
$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) = \frac{\Gamma(d/2)}{2(2\pi)^d} \int_0^t \frac{1}{s^{d/2}} ds,$$

which forces,

$$d/2 < 1 \quad \Longleftrightarrow \quad d < 2.$$

- Thus a global space-time white noise solution may only exist when the spatial dimension is 1.

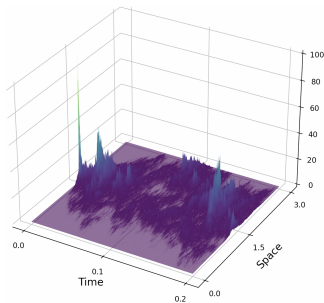
$$u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W} \text{ with } u_0(x) = \delta_0(x) \quad (\text{white noise})$$



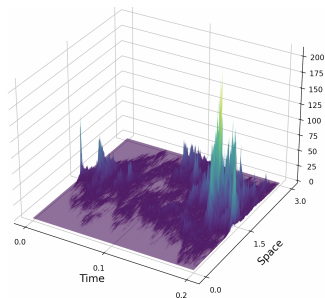
- Simulations done by using a finite difference scheme [5].
- We partition (space) $[0, 3]$ into n_x pieces of size $\delta_x = 3/n_x$.
- We partition (time) $[0, .2]$ into n_t pieces of size $\delta_t = .2/n_t$.
- To control error, we assume $n_x^2 \delta_t < \frac{1}{2}$ which implies $n_t \gg n_x$.

$$u(t + \delta_t, x) = u(t, x) + n_x^2 \delta_t (u(t, x + \delta_x) + u(t, x - \delta_x) - 2u(t, x)) \\ + n_x \lambda u(t, x) \text{normal}(0, \delta_t \delta_x).$$

Intermittency of $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ (white noise)



$\lambda = 1.8$



$\lambda = 2$

● Notice the enlargement of the peaks as λ increases. We have the following theorem.

Theorem (Chen and Dalang [1, Theorem 2.4])

The second moments of the stochastic heat equation driven by space-time white noise grow exponentially in time.

Will the second moments ever be bounded in time?

- Several slides below will require the second moments to be bounded in time:

$$\sup_{t \geq 0} \|u(t, x)\|_2 < \infty, \quad \text{where } \|X\|_p = [\mathbb{E}(|X|^p)]^{1/p}.$$

- This is possible, however this will require us to only consider $d \geq 3$, and therefore we must only consider a spatially colored noise.
- Recall from above, that the covariance of a spatially colored noise is given by the following:

$$\begin{aligned} \mathbb{E}[W(\phi)W(\psi)] &= \int_0^\infty ds \int_{\mathbb{R}^d} dx f(x)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x) \\ &= \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x) f(x - y) \psi(s, y) \\ &= (2\pi)^{-d} \int_0^\infty ds \int_{\mathbb{R}^d} d\xi \hat{f}(\xi) \mathcal{F}\phi(s, \xi) \overline{\mathcal{F}\psi(s, \xi)} \end{aligned}$$

$$u_t - \frac{1}{2} u_{xx} = b(u) \dot{W} \text{ with } u(0, \cdot) = \mu(\cdot) \quad (\text{colored noise})$$

Theorem (Chen and Huang [2, Theorem 1.2])

There exists a unique global mild solution under the following:

- $b : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ such that there exists a $L_b, L_0 > 0$ so for all $u, v \in \mathbb{R}$ and $x \in \mathbb{R}^d$,

$$|b(x, u) - b(x, v)| \leq L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0.$$

- The initial measure, $\mu(dx)$ (resp. $\varphi(x)dx$), satisfies for some $a > 0$,

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\mu|(dx) < \infty \quad \left(\text{resp. } \int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx < \infty \right),$$

where $|\mu| = \mu^+ + \mu^-$ is the *Hahn decomposition*.

- The spectral density, $\hat{f}$, satisfies *Dalang's condition* for some $\beta > 0$:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

Bounded moments in time

Theorem (Chen and Kim [4, Theorem 1.3 and Lemma 2.5])

Suppose that we strengthen Dalang's condition and the Lipschitz assumption to the following:

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty \quad \text{and} \quad L_b^2 < \frac{1}{128\Upsilon(0)}.$$

Then the following holds:

$$\begin{aligned} \|u(t, x)\|_2 &\leq C_2 \left[1 + \left(G(t, \cdot) * |\mu| \right)(x) \right] \\ &= C_2 \left[1 + \int_{\mathbb{R}^d} G(t, x - y) |\mu|(dy) \right], \quad \text{for all } t \geq 0. \end{aligned}$$

- If $\mu = \delta_0$ then $\|u(t, x)\|_2 \leq C_2 [1 + (2\pi t)^{-d/2}]$.
- Later, it will suffice consider μ such that $\sup_{t \geq t_0} \|u(t, x)\|_2 < \infty$.

Transition functions and their semigroup

Definition (Markovian Transition Functions)

- Let H be a Hilbert space and $\mathcal{B}(H)$ be the borel sets of H .
- $\{P_t(x, \Gamma) : t \geq 0, x \in H, \Gamma \in \mathcal{B}(H)\}$ forms a family of *Markovian transition functions* if,
 - $x \mapsto P_t(x, \Gamma)$ if measurable,
 - $\Gamma \mapsto P_t(x, \Gamma)$ is a probability measure on H ,
 - $P_{t+s}(x, \Gamma) = \int_H P_t(x, dy) P_s(y, \Gamma)$,
 - $P_0(x, \Gamma) = \mathbf{1}_\Gamma(x)$.

Definition (Transition Semigroup)

The above transition functions form the following semi group on $\mathcal{B}(H)$, which denotes the bounded Borel measurable functions on H :

$$P_t \varphi(x) = \int_H \varphi(y) P_t(x, dy), \quad \varphi \in \mathcal{B}(H).$$

Invariant measures

Definition (Invariant measures)

- A probability measure, η , on H is said to be *invariant* for the transition functions P_t if for all $\Gamma \in \mathcal{B}(H)$ and $t > 0$,

$$P_t^*(\eta)(\Gamma) := \int_H P_t(x, \Gamma) \eta(dx) = \int_H P_0(x, \Gamma) \eta(dx) = \int_H \mathbf{1}_\Gamma(x) \eta(dx) = \eta(\Gamma),$$

which is equivalent to,

$$\int_H P_t \varphi(x) \eta(dx) = \int_H \varphi(x) \eta(dx), \quad \text{for all } \varphi \in \mathcal{B}(H).$$

- This implies that the η -average of $x \mapsto P_t(x, \Gamma)$ is $\eta(\Gamma)$ and this is independent of time:

$$\text{AVG}_\eta[P_t(\cdot, \Gamma)] = \frac{1}{\eta(H)} \int_H P_t(x, \Gamma) \eta(dx) = \int_H P_t(x, \Gamma) \eta(dx) = \eta(\Gamma).$$

Invariant measures for the stochastic heat equation

- First we define the law of the solution as follows:

$$\mathcal{L}(u(t, \cdot; \zeta))(\Gamma) = \mathbb{P}[\omega \in \Omega : u(t, \cdot; \zeta) \in \Gamma], \quad \text{for all } \zeta \in H \text{ and } \Gamma \in \mathcal{B}(H),$$

and for the mean time we are intentionally vague about H .

- Our intention is for

$$P_t(\zeta, \Gamma) := \mathcal{L}(u(t, \cdot; \zeta))(\Gamma)$$

to form a Markovian semigroup.

- Established theory given by DaPrato and Zabczyk [7] requires our solution, specifically $\zeta \mapsto u(t, \cdot; \zeta)$, to be almost surely H -valued for $t > 0$, otherwise $\mathcal{L}(u(t, \cdot; \zeta))(\Gamma)$ would not have too much meaning.
- So what is H ?

$H = L^2_\rho(\mathbb{R}^d)$: weighted square integrable functions

Definition (Admissible Weight)

A positive, bounded and continuous $L^1(\mathbb{R}^d)$ function, ρ , is an *admissible weight* if

$$\left(G(t, \cdot) * \rho(\cdot) \right)(x) \leq C_\rho \rho(x), \quad \text{for all } t \in [0, T].$$

Definition ($L^2_\rho(\mathbb{R}^d)$)

The space $H := L^2_\rho(\mathbb{R}^d)$ is the space of ρ -weighted square integrable functions:

$$\|\zeta\|_\rho^2 := \int_{\mathbb{R}^d} |\zeta(x)|^2 \rho(x) dx < \infty.$$

Moreover, H is a Hilbert space with inner product defined as

$$\langle \zeta, \psi \rangle_\rho := \int_{\mathbb{R}^d} \zeta(x) \psi(x) \rho(x) dx.$$

Weight examples

- The following two are examples of admissible weights:

$$\exp(-a|x|), \text{ for } a > 0 \quad \text{and} \quad \frac{1}{1 + |x|^a}, \text{ for } a > d.$$

- One can show that the following is not admissible for any $a > 0$,

$$\exp(-a|x|^2),$$

- Numerically it is suggested that for any $1 < b < 2$ and $a > 0$ that

$$\exp(-a|x|^b),$$

is not admissible, but this has not been proven.

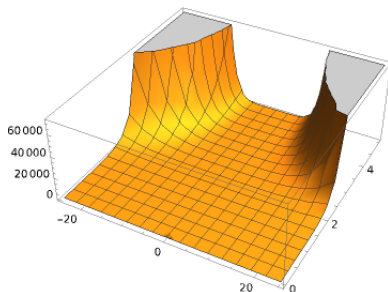


Figure: Plot of $\frac{(G(t,\cdot) * \exp(-|\cdot|^{1.25}))(x)}{\exp(-|x|^{1.25})}$

The solution is almost surely H -valued.

Theorem

Let ρ be an admissible weight and suppose that

$$\Upsilon(0) = \int_{\mathbb{R}^d} \frac{\widehat{f}(\xi)}{|\xi|^2} d\xi < \infty \quad \text{and} \quad L_b^2 < \frac{1}{128\Upsilon(0)}.$$

Now suppose our initial measure, μ , satisfies the following:

$$\mathcal{G}_\rho(t; \mu) := \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x - y) |\mu|(dy) \right)^2 \rho(x) dx < \infty.$$

Then for all $t > 0$,

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t; \mu) < \infty.$$

References



L. Chen and R. Dalang (2015)

Moments and growth indices for the nonlinear stochastic heat equation with rough initial conditions

The Annals of Probability 43, 6, 3006–3051



L. Chen and J. Huang

Comparison principle for stochastic heat equation on $\mathbb{R}^d$

Ann. Probab., 47.2 (2019) 989–1035.



L. Chen, Y. Hu, D. Naulart (2019)

Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$.

Stochastic Processes and their Applications, 129 5073–5112.



L. Chen and K. Kim (2019)

Nonlinear stochastic heat equation driven by a spatially colored noise: moments and intermittency.

Acta Mathematica Scientia 39B(3), 645–668



A. Davie and J. Gaines (2000)

Convergence of numerical schemes for the solution of parabolic stochastic